

N067AV (S/N: W20051876422) — NVMe SSD Hardware Failure Report

Immfly Ground Operations / IT Session: September 15, 2026

Field	Value
Machine	N067AV (S/N: W20051876422)
Affected component	ADATA IM2P33E8-001TD 1TB NVMe Solid State Drive (/dev/nvme0n1)
Initial symptom	System boot failure. System drops into emergency maintenance mode with repeated kernel I/O aborts on device nvme0n1p5 during startup filesystem check.
Hardware status	HARDWARE FAILURE — Critical medium read error on sectors 791664640+, 1143986176, 1357895680, 1571805184. ECC unrecoverable.
Root cause	Severe physical NAND flash cell degradation and exhaustion of reserve spare blocks on the ADATA NVMe SSD. The storage controller returns critical media errors to the kernel block layer, rendering /mnt/wwwdata unmountable.
Comparison	Distinct from fleet modem failures (N109AV, N136AV, N060AV). This is an irrecoverable primary storage device breakdown halting the Linux OS bootloader.
Recommended action	Physical NVMe SSD replacement required. Replace SSD unit, re-image operating system, and deploy content packages.

NARRATIVE SUMMARY

During regular server startup, aircraft server unit **N067AV** suffered a catastrophic boot failure. While early boot services, kernel modules, swap, and minor partitions (/boot/efi and /kvmroot) mounted successfully, the system halted during the mandatory filesystem check of partition nvme0n1p5 (UUID ...b717-4301-8ad3a9ccb6c7d854), designated for /mnt/wwwdata.

The Linux kernel block layer reported persistent blk_update_request: critical medium error dev nvme0n1 alongside multiple cascading Buffer I/O error on dev nvme0n1p5, logical block 591396952-591396959, async page read failures. In the NVMe protocol specification, a critical medium error indicates physical NAND flash cell exhaustion where the internal ECC engine can no longer reconstruct data nor reallocate blocks.

Because /mnt/wwwdata is defined as a critical dependency for local filesystems, systemd-fsck failed with an exit error, triggering an immediate drop to the Emergency Maintenance Shell (Give root password for maintenance). Software-level disk repair (fsck) cannot rectify physical NAND cell breakdown confirming permanent hardware failure of the storage drive.

DIAGNOSTIC FINDINGS & SEQUENCE OF EVENTS

- 1. **Early boot initialization:**
Kernel loaded core modules, remounted root, activated swap, and enumerated the primary PCIe drive as ADATA_IM2P33E8-001TD.
- 2. **Secondary partition verification:**
Filesystem checks passed successfully for /boot/efi (FAT) and /kvmroot (ext4), and both mounts completed without incident.
- 3. **Physical medium read failure on nvme0n1:**
At 70% of the fsck pass on partition 5, the NVMe driver received unrecoverable read faults across multiple sector ranges (791664640, 1143986176, 1357895680, 1571805184).
- 4. **Buffer I/O page read exhaustion:**
Sequential reads on logical blocks 591396952 through 591396959 failed repeatedly with async page read errors, indicating physical damage to the partition superblocks and metadata tables.
- 5. **Systemd boot target failure:**
Service systemd-fsck@dev-disk-by\x2duuid-...a9ccb6c7d854.service exited with failure. Dependencies for /mnt/wwwdata and Local File Systems collapsed.
- 6. **Emergency Shell lockdown:**
Host aborted startup targets and prompted for root maintenance credentials. The unit is out of service.

CONCLUSION AND RECOMMENDATION

The ADATA IM2P33E8 1TB NVMe drive in server **N067AV** has suffered unrecoverable physical hardware degradation. The storage device cannot maintain data integrity, and attempting software repair or sector remapping is futile due to depleted NAND spare capacity.

Recommended action: Issue an urgent Line Maintenance Work Order to **physically replace the NVMe SSI (ADATA 1TB M.2)** in aircraft **N067AV (S/N: W20051876422)**, followed by standard base image provisioning and multimedia content reload.

APPENDIX: CONSOLE BOOT FAILURE OUTPUT

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[BOOT:N067AV] Kernel block layer error & filesystem check failure:
[ 10.153265] systemd-journald[401]: Received client request to flush runtime journal.
[ OK ] Found device ADATA_IM2P33E8-001TD EFI.
[ OK ] Found device ADATA_IM2P33E8-001TD kvmroot.
[ OK ] Found device ADATA_IM2P33E8-001TD wwwdata.
[ OK ] Mounted /kvmroot.
[ OK ] Mounted /boot/efi.
Checking in progress on 1 disk (70.0% complete)
[ 12.294361] blk_update_request: critical medium error, dev nvme0n1, sector 791664640 op 0x0:(READ) flags 0x80700 phys_seg 1 prio class 0
[ 12.307268] blk_update_request: critical medium error, dev nvme0n1, sector 1143986176 op 0x0:(READ) flags 0x80700 phys_seg 1 prio class 0
[ 12.319620] blk_update_request: critical medium error, dev nvme0n1, sector 1357895680 op 0x0:(READ) flags 0x80700 phys_seg 3 prio class 0
[ 12.335298] blk_update_request: critical medium error, dev nvme0n1, sector 1571805184 op 0x0:(READ) flags 0x80700 phys_seg 2 prio class 0
[ 17.363777] blk_update_request: critical medium error, dev nvme0n1, sector 791664728 op 0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[ 17.376001] Buffer I/O error on dev nvme0n1, logical block 591396952, async page read
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[ 17.370009] Buffer I/O error on dev nvme0n1p5, logical block 591396952, async page read
[ 17.384014] blk_update_request: critical medium error, dev nvme0n1, sector 791664729 op 0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[ 17.395915] Buffer I/O error on dev nvme0n1p5, logical block 591396953, async page read
[ 17.403915] blk_update_request: critical medium error, dev nvme0n1, sector 791664730 op 0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[ 17.415813] Buffer I/O error on dev nvme0n1p5, logical block 591396954, async page read
[ 17.567052] blk_update_request: critical medium error, dev nvme0n1, sector 791664728 op 0x0:(READ) flags 0x0 phys_seg 1 prio class 0
[ 17.579590] Buffer I/O error on dev nvme0n1p5, logical block 591396952, async page read
[ OK ] Started Emergency Shell.
[FAILED] Failed to start File System Check...c-b717-4301-8ad3-a9ccb6c7d854.
See 'systemctl status "systemd-fsck@dev...x2da9ccb6c7d854.service"' for details.
[DEPEND] Dependency failed for /mnt/wwwdata.
[DEPEND] Dependency failed for Local File Systems.
You are in emergency mode. After logging in, type Give root password for maintenance
(or press Control-D to continue):
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